

Algebraic Geometry Princeton

Quals Questions

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Question 0.0. (Kollar) *Compute the genus of the curve $x^3 = y^6 + 1$*

Question 0.1. (Katz) *Define the genus of a curve in every way that you know how. Why are these equal?*

Question 0.2. (Katz) *State Serre Duality. Define the Hilbert Polynomial.*

Question 0.3. (Katz) *Compute the genus of the Fermat curve in every way you can.*

Question 0.4. (Katz) *What does adjunction say in the case of the Fermat curve? How about a degree d hypersurface?*

Question 0.5. (Katz) *What are such hypersurfaces called (K3 surfaces)?*

Question 0.6. (Katz) *What happens if I know the number of points of the variety over a finite field with p^r elements for all r ?*

Question 0.7. (Katz) *Suppose a variety has exactly $p^n + 1$ points over a finite field of characteristic p^n . Specifically, consider the projective n -space, P^n , over a finite field with p^n elements.*

What can we deduce about this variety?

Do you know how to prove this?

Question 0.8. (Katz) *State the Weil Conjectures*

Question 0.9. (Katz) *Suppose that my equations are defined over $\mathbb{Z}$. What can I say when I base change to various fields?*

Question 0.10. (Katz) *State Riemann-Roch. What is Riemann Roch good for?*

Question 0.11. (Katz) *What's the relationship between divisors and sections of invertible sheaves?*

Question 0.12. (Katz) *What would Weierstrass say about elliptic curves?*

Question 0.13. (Katz) *Define the Weierstrass p -function. What are its basic properties? What can you say about its poles? Does it satisfy a differential equation?*

How would you know that there should exist such a differential equation?

Question 0.14. (Katz) *Give me a 30 second sketch of everything you know about Riemann-Roch in higher dimensions. What is this useful for?*

Question 0.15. (Bhatt) *Embed a genus 1 curve in projective space.*

Question 0.16. (Venkatesh) *Why is it called an elliptic curve?*

Question 0.17. (Bhatt) *What if it's over $\mathbb{R}$ and has no $\mathbb{R}$ -points? Then can you embed in $\mathbb{P}^2$? What about a finite field?*

Question 0.18. (Bhatt) *How can you always find an $\mathbb{F}_p$ point of a genus 1 curve?*

Question 0.19. (Bhatt) *Can I embed a genus 1 curve in higher P^N as a complete intersection?*

Question 0.20. (Bhatt) *Give an example of a smooth affine curve which has nontrivial Picard group.*

Question 0.21. (Bhatt) *Can $\text{Pic}(X)$ be finitely generated? Work over $\mathbb{C}$ and countable field cases.*

Question 0.22. (Bhatt)

Question 0.23. (Bhatt) *If I blow up X surface at a point how does the cohomology change?*

Question 0.24. (Bhatt)

Question 0.25. (Bhatt) *1*

Question 0.26. (Faltings) *What is a scheme?*

Question 0.27. (Faltings) Define the genus of a curve (in a couple of ways). Explain how to compute it using the Hilbert polynomial. Why is this the same as the dimension of the space of global 1-forms?

Question 0.28. (Faltings) For a coherent sheaf F on $\mathbb{P}^r$ associated to the graded module M , why is the Hilbert polynomial $H(n)$ ($:=$ Euler characteristic of $F(n)$) equal to the dimension of M_n for $n \gg 0$?

Question 0.29. (Faltings) Compute the genus of the curve $y^2 = x^5 - x$ by using Hurwitz' formula.

Question 0.30. (Xu, Chengyang) What do you know about minimal model program on rational surfaces. Why are they different? How do you show they are connected by blow-up and blow-down?

Question 0.31. (Xu, Chengyang) What is the class of the canonical divisor on $\mathbb{F}_n$?

Question 0.32. (Xu, Chengyang) In general, consider a ruled surface $f : X = \mathbb{P}(E) \rightarrow B$ on a curve B of higher genus, E is a rank 2 vector bundle, what is the class of the canonical divisor.

Question 0.33. (Xu, Chengyang) What is the class group of a smooth affine conic curve (say, in characteristic 0).

Question 0.34. (de Jong) What is an elliptic curve?

Question 0.35. (de Jong) Write down a cubic plane curve. Where is the identity? Define the group law for this curve. How do you define it geometrically?

Question 0.36. (de Jong) How do you define the group law for a nodal cubic plane curve? How do you do it geometrically?

Question 0.37. (de Jong) How would you define a family of curves? Why do you want flatness?

Question 0.38. (de Jong) Suppose you have a not-necessarily flat family over a base scheme. How would you impose conditions to make it a flat family?

Question 0.39. (de Jong) Take a quadric and a cubic hypersurface in $\mathbb{P}^3$ and intersect them. Suppose the resulting curve is as nice as you want. What is its genus?

Question 0.40. (de Jong) State the adjunction formula.

Question 0.41. (de Jong) Define genus. What is the canonical sheaf?

Question 0.42. (Kollar) *Compute the genus of $y^3 = x^6 - 1$*

Question 0.43. (Kollar) *If X is a projective scheme over the rationals $\mathbb{Q}$, K a finite extension of $\mathbb{Q}$, and X' the base change of X to K , can you relate the cohomology of the structure sheaf of X' to the cohomology of the structure sheaf of X ? What if we just look at H^0 ? How about for H^1 and if we let $X = \text{Spec } \mathbb{Q}[x]/(x^2+1)$ and K a field containing $\mathbb{Q}[i]$?*

Question 0.44. (Faltings) *Can a genus 2 curve embed the plane? Prove what you need*

Question 0.45. (Faltings) *Give a lower bound for the degree of a map from a genus g curve to $\mathbb{P}^1$. Can there be an upper bound?*

Question 0.46. (Faltings) *How would you write down all the holomorphic differentials on the Fermat curve?*

Question 0.47. (Faltings) *Give an example of a non-hyperelliptic curve. Explain.*

Question 0.48. (Faltings) *What is the Jacobian of a curve? How do you determine its dimension?*

Question 0.49. (Faltings) *What is the theorem of the cube? What is it used for?*

Question 0.50. (Faltings) *Compute the cohomology of projective space.*

Question 0.51. (Faltings) *Does every curve embed $\mathbb{P}^3$? Why are all curves even projective?*

What about embedding a curve into $\mathbb{P}^2$?

Question 0.52. (Faltings) *What is a regular scheme? Is regularity the same as smoothness over any base field?*

Question 0.53. (Faltings) *Do you know what Neron minimal models are? Determinant of cohomology?*

Question 0.54. (Kollar) *Define arithmetic genus. Prove the genus formula. State the explicit computation giving $H^i(\mathbb{P}^n, \mathcal{O}(m))$ and Serre duality*

Question 0.55. (Shou-Wu) *Tell me the definition of a scheme, give examples of affine scheme, projective scheme and a scheme that is not affine nor projective.*

Question 0.56. (Shou-Wu) *Why is $X = \mathbb{A}^2 - \{0\}$ not projective nor affine? what does the cohomology of affine/projective scheme look like?*

Question 0.57. (Shou-Wu) *So we have a map f from X to $\mathbb{P}^2$, there is a G_m action on X , f is an affine map? The push forward of the structure sheaf of X has a decomposition?*

Question 0.58. (Shou-Wu) *Define curves (smooth) and genus. Classify the curves of genus 0, 1, 2, 3*

Question 0.59. (Tang) *Why is the elliptic curve of degree 3?*

Question 0.60. (Tang) *Give the map from elliptic curve to $\mathbb{P}^2$. How do you find that line bundle?*

Question 0.61. (Boyu) *what is the definition of hyperelliptic curves*

Question 0.62. (Boyu) *So is there a meromorphic function from a curve of lower genus to higher genus?*

Question 0.63. (Shou-Wu) *compute the genus of the curve $y^5 = x(x - 1)(x - \lambda)$*

Question 0.64. (Katz) *State Riemann-Roch for curves and Riemann Surfaces*

Question 0.65. (Katz) *Define the arithmetic genus and geometric genus of a variety.*

Question 0.66. (Katz) *What can you say about global sections of $\Gamma(X, \omega)$? Sections of $H^1(X, \mathcal{O})$?*

Question 0.67. (Katz) *What is the genus of $y^2 = x^{11} - 1$? How about $y^2 = f(x)$ with $f(x)$ of odd degree and distinct roots? Give the general genus formula for that curve*

Question 0.68. (Katz) *Talk about intersection theory for a surface.*

Question 0.69. (Katz) *Why does an elliptic curve have a cubic form in $\mathbb{P}^2$?*

Question 0.70. (Katz) *State Abel's Theorem for an elliptic curve. What about higher dimensional varieties?*

Question 0.71. (Katz) *What does Hasse's Theorem say about the number of points on an elliptic curve over a finite field.*

Question 0.72. (Katz) *How does this relate to the Weil Zeta Function of a projective variety? What does this zeta function look like for an elliptic curve? What does it look like for $\mathbb{P}^n$?*

(Hint: Identical Denominator)

Question 0.73. (Katz) *Where does the coefficient a_p come from in the elliptic curve's Zeta function?*

Question 0.74. (Katz) *Suppose that you have a large finite set of equations over a finite field and the number of points on the resulting variety over $\mathbb{F}_{q^n}$ for all n . Further suppose that you know that the resulting variety is a hypersurface. How might one determine the genus of this hypersurface?*

(Hint: Look at the numerator of some zeta function.)

Question 0.75. (Katz) *Why does an elliptic curve have a cubic form in P^2 ?*

Question 0.76. (Katz) *Why does an elliptic curve have a cubic form in P^2 ?*

1 Notes

Careful with Shonwu, pretty recent